

OCR Further Mathematics
Statistics - Linear Regression
Mark Scheme
AS and A Level
Further Mathematics A - H235, H245



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Question			Answer/Indicative content	Marks	Part marks and guidance	
1	a		$y = 52.7 + 0.251x$	B1* (AO1.1) B1* (AO1.1) depB1 (AO1.1) [3]	a in range [0.250, 0.251] b correct to 3 SF Completely correct including letters SC: Correct formulae used for a and b : M1(A1)A1	
	b		This quantity is minimised to find best-fit line	B1 (AO2.4) [1]	Need “minimised” or “this is its minimum value” OE	
	c		$y' = 11.5 + 0.139x$ $[y' = \frac{5}{9} \times (\text{their } a - 32) + \frac{5}{9} \times \text{their } b]$	M1 (AO1.1) A1ft (AO1.1) [2]	Apply inverse formula at least once All correct, any letters, ft on their y	
			Total	6		
2	a		Vertical lines drawn from points Best-fit line minimises squares of lengths of these lines	M1 (AO1.1) A1 (AO2.4) [2]	Or squares with correct sides oe, clearly stated	
	b	i	$u = -3.55 + 2.05t$	M1 (AO1.1) A1 (AO1.1) A1 (AO1.1) [3]	Evidence for correct method for b (e.g. right answer) Both numbers right All correct including letters	
		ii	$w - 4 = -3.55 + 2.05t/2$ $w = 0.45 + 1.025v$	M1 (AO3.1a) A1 (AO1.1) [2]	Put $t = v/2$, $u = w - 4$ Fully correct, simplified to 3 terms	
			Total	7		
3	a		Neither	B1 (AO1.2) [1]		



Question			Answer/Indicative content	Marks	Part marks and guidance	
	b		$q = 1.13 + 0.620p$	B1B1 (AO1.1) (AO1.1) B1 (AO1.1) [3]	0.62(0) correct; both numbers correct Fully correct answer including letters	
	c	i	2.68	B1ft (AO1.1) [1]	awrt 2.68, ft on their (b) if letters correct	
		ii	2.5 is within data range, and points (here) are close to line / well correlated	B1 (AO2.2b) B1 (AO2.2b) [2]	At least one reason, allow "no because points not close to line" Full argument, two reasons needed	
	d		Not much data here / points scattered / possible outliers So not very reliable	M1 (AO2.3) A1 (AO1.1) [2]	Reason for not very reliable (<i>not</i> "extrapolation") Full argument and conclusion, not too assertive (not wholly unreliable!)	
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance		
4	a		0.8392	B1(AO1.1) [1]	Awrt 0.839 $S_{xx} = 1.7449\dots$ $S_{yy} = 41.2\dots$ $S_{xy} = 7.116\dots$ <u>Examiner's Comments</u> For most candidates this was a very straightforward start to the paper.		
	b		$y = -1.180 + 4.0781x$	B1(AO1.1) B1(AO1.1) [2]	Both coeffs, awrt -1.18 and 4.08 Letters correct, needs 1 correct coefficient <u>Examiner's Comments</u> Correct answers were very common here, although a few candidates used their product-moment correlation coefficient as a. The variable letters x and y were almost always correctly given.		
	c		Value of PMCC suggests that there is strong correlation, or 0.75 <u>shown</u> close to mean 0.399	B1(AO3.5a) [1]	E.g. "r high so points lie close to line". "r is high" alone is enough. No wrong extras <u>Examiner's Comments</u> Many candidates produced unnecessarily complicated answers. There are only two standard criteria to consider for the reliability of an estimate, correlation and	Not "0.75 is close to mean", unless properly justified, e.g. SD (= 0.264) calculated	



Question			Answer/Indicative content	Marks	Part marks and guidance	
					whether the given data value (here 0.75) is in the data range. As candidates should have got in Question 1(a) a value of r that is high, there was no reason for them to respond with an answer any more complicated than 'r is close to 1 so there is strong correlation'. Those who attempted to use the other condition needed to justify that 0.75 was likely to be within the range of the data (for instance by finding both the sample mean (0.399) and the standard deviation), but this cannot be known with certainty from the information given. Exemplar 1 10) The estimate will be reliable when $n = 0.75$ because n is a large number of 25 so a large sample is used to find the regression line so anomalies will not affect it too much. This is a typical irrelevant answer. Exemplar 2 10) The average value of x is 0.399 and $y = 0.75$ is close to this so this is a reasonable value. This assertion is not valid without further calculation, for example of the standard deviation. The deleted words make no difference.	



Question			Answer/Indicative content	Marks	Part marks and guidance		
	d		Whether $x = 0.75$ is within the data range	B1(AO 3.5b) [1]	E.g. “maximum and minimum values of x ”; not “all data points”. No wrong extras	Or clear reference to interpolation. NB: 95% CI for x is (-0.156, 0.954)	
					<u>Examiner's Comments</u> Following from the comments on part (c), it should be clear that all that was needed here was ‘the range of values of x ’. It is not necessary to know all the data values and those who said that they need to know the correct corresponding value of y were missing the point. Some said that the data had to be bivariate normal, or elliptically distributed, but this is not necessary unless a formal hypothesis test is being carried out. About half of the candidates picked up the mark here.		
			Total	5			



Question			Answer/Indicative content	Marks	Part marks and guidance		
5		a	Controlled variable, independent variable	B1 (AO 2.5) [1]	Any extras or omissions: B0		
		b	Draw vertical line from point to regression line; state the Least Squares Regression Line minimises sum of squares of these lengths	B1 (AO 1.2) E1 (AO 2.4) [2]	Allow this mark from horizontal lines		
		c	Both coefficients divided by 2.54	B1 (AO 1.1) [1]	oe		
			Total	4			

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Question			Answer/Indicative content	Marks	Part marks and guidance		
6		a	Neither	B1(AO2.5) [1]	OE	Not “neither is independent of the other”	
					<u>Examiner’s Comments</u> In this question neither variable was independent. Many thought that c was independent so that m depended on it, but c is not controlled; this is a genuine bivariate situation.		
		b	$c = 2.848 - 0.1567m$ BC	B1(AO1.1) B1(AO1.1) B1(AO1.1) [3]	Correct a , awrt 2.85 Correct b , awrt 0.157 Letters correct from correct method (If both wrongly rounded, e.g. $c = 2.84 - 0.156m$, give B2)	SC: m on c : $m = 15.65 - 4.832c$: B2 $y = 15.65 - 4.832x$: B1 $c = 15.65 - 4.832m$: B1 If B0B0, give B1 for correct letters from valid working	
					<u>Examiner’s Comments</u> Some candidates found the equation of the least squares regression line of m on c , rather than c on m . Consequently the answer $m = 15.7 - 4.83c$ was seen almost as often as the correct answer $c = 2.85 - 0.157m$. Many worked with x and y , so that $y = 15.7 - 4.83x$ was also commonly seen.		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		c	a unchanged, b multiplied by 2.2 (allow “ a unchanged, b increases”, etc)	B1(AO2.2a) [1]	oe, e.g. $c = 2.848 - 0.345m$; $m = 7.114 - 2.196c$ SC: m on c in (b): Both divided by 2.2 B1	
<u>Examiner's Comments</u> Few correct responses were seen here, even though responses from part (b) were followed through. Many thought that the coefficients were unchanged by linear coding. Others could not work out whether the coefficients had to be divided or multiplied by 2.2.						

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Question			Answer/Indicative content	Marks	Part marks and guidance		
		d	Draw approximate line of best fit Draw at least one vertical from line to point Say that "Best fit" line minimises the sum of squares of these distances	M1(AO1.1) M1(AO2.4) A1(AO2.4)[3]	Needs M2 and "minimises" and "sums of squares" or <u>Examiner's Comments</u> Candidates needed to draw vertical (or horizontal) displacements from an approximate line of best fit and then say that the sum of the squares of these displacements was minimised. Full credit could not be given for vague statements such as "least squares gives the line nearest to the points". A common incorrect response was to draw diagonal lines from the points to their line.	SC: Horizontal(s): full marks (indept of (b))	
			Total	8			

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Question		Answer/Indicative content	Marks	Part marks and guidance		
7		$S_{xx} = \frac{42767}{50} - \left(\frac{1270}{50}\right)^2 = 210.18$ $S_{xy} = \frac{173161}{50} - \frac{1270}{50} \times \frac{5173}{50} = 835.336$ $b = S_{xy}/S_{xx} = 3.97438...$ $a = 103.46 - 25.4 \times b = 2.510...$ $y = 2.510 + 20 \times 3.974$ $= 82.0$ $\hat{y} \pm 1.96 \sqrt{\frac{1.9}{10}}$ $= (81.14, 82.85)$	M1 (AO 2.1) A1 (AO 1.1) A1 (AO 1.1) M1 (AO 3.3) A1 (AO 1.1) M1 (AO 1.1a) A1 (AO 1.1a) A1 (AO 2.2a) [8]	Correct method for b Correct value of b, in [3.97, 3.98] Correct value of a, awrt 2.51 Estimate y Correct value $y \pm z\sqrt{\sigma^2/n}$ $z = 1.96$ Both correct, awrt 4 sf. Allow [81.1, 81.2] and [82.8, 82.9] only if all other numbers fully correct		
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Question			Answer/Indicative content	Marks	Part marks and guidance		
8		a	t is a controlled variable	B1(AO 2.5) [1]			
		b	0.987	B1(AO 1.2) [1]	Or “unchanged”		
		c	1	B1(AO 2.2a) [1]			
		d	Points lie close to a straight line	B1(AO 2.2b) [1]	Not “positive gradient” – that information is given by t/s		
			Total	4			



Question			Answer/Indicative content	Marks	Part marks and guidance		
9		a	N is likely to be dependent on T because more ice creams are likely to be sold in hot weather	B1(AO2.5) [1]	oe, any plausible reason that shows understanding of concept		
		b	$r = 0.819$	B2(AO1.1b)(AO1.2) [2]	BC: awrt 0.819 Correct answer, no working: B2	SC: answer incorrect, correct working seen: B1	
		c	No change, r unchanged by linear coding	B1(AO2.3) [1]	Both "no change" oe and "linear" oe		
		d	$n = -15.6 + 1.81t$	B1ft(AO1.1b)depB1(AO1.1b) [2]	BC, Numbers correct, to 3 sf Whole equation correct		
		e	Vertical line drawn between line and a point Minimise sum of squares of these lengths	M1*(AO2.4) depE1(AO2.4) [2]	At least one line drawn This or equivalent stated		
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Question			Answer/Indicative content	Marks	Part marks and guidance		
10		a	Independent and controlled	B1(AO1.2) [1]	Both, no others		
		b	$d = 1.61v - 24.1$	B2(AO1.1) (AO3.3) [2]	All correct including letters, 3 s.f. BC B1 Numbers right but not letters	$[d = 1.614v - 24.143]$	
		c	$d = 1.61 \times 45 - 24.1 = 48$ to the nearest whole number	B1(AO3.4) [1]	awrt 48.5		
		d	Yes as r is close to 1 and 45 is within data range	E1(AO3.5a) E1(AO3.5b) [2]	Yes with one reason Second reason		
			Total	6			



Question			Answer/Indicative content	Marks	Part marks and guidance		
11		a	$y = 157x - 988$	B1(AO1.1) B1(AO3.3) [2]	Either coefficient correct to 3 s.f. All correct including letters	BC [$y = 157.11x - 988.15$]	
		b	No; r is unaffected by linear coding	E1(AO2.3) [1]	Needs both “no” and “linear” seen, and no wrong or irrelevant statements		
		c	(i) $H_0: \rho = 0$ $H_1: \rho \neq 0$ where ρ is the population correlation Coefficient CV 0.5494 $0.9586 > 0.5494$ Reject H_0 Significant evidence of correlation between consumption of mozzarella cheese and number of civil engineering doctorates awarded.	B2(AO1.1) (AO2.5) B1(AO3.4) M1(AO1.1) M1(AO1.1) A1(AO2.2b) [6]	Hypotheses correctly stated B1 if no symbol used Correct critical value seen Explicit comparison Correct first conclusion Contextualised, acknowledged uncertainty		



Question			Answer/Indicative content	Marks	Part marks and guidance		
		c	(ii) No, correlation does not imply causality	E1(AO3.5b) [1]	Or: "there may be another factor that influences both"		
			Total	10			



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Question			Answer/Indicative content	Marks	Part marks and guidance	
12		i	$S_{xx} = 503.45 - \frac{70.3^2}{10}$ (= 9.241) $S_{yy} = 103.94 - \frac{30.8^2}{10}$ (= 9.076)			
		i	$S_{xy} = 211.9 - \frac{70.3 \times 30.8}{10}$ (= -4.624)	M1	Correct sub in any correct S formula	Must be correct sub in all S 's & r but not nec'y accurate
		i	= -0.5049.... or -0.505 (3 sfs)	M1	Correct sub in any correct r formula	
		i	= -0.5049.... or -0.505 (3 sfs)	A1	Correct ans with no wking: M1M1A1	
					<u>Examiner's Comments</u> This part was well answered. A few candidates made errors such as those mentioned above. Some candidates used the statistical functions on their calculator and gave an answer with no working. As noted above, this is a risky strategy, resulting in either full marks or no marks. Candidates who wish to use this method should be advised to carry out the whole calculation a second time, as a check. The safest approach is to show the substitutions into the relevant formulae for S_{xy} etc and for b and r .	

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	Correlation (of UR & CPI) does not imply causation or r not close to -1	B1	Both (UR & CPI) may depend on another factor or r small or poor corr'n or ignore all else <u>Examiner's Comments</u> A good number of candidates answered correctly either by pointing out that correlation does not imply causation or that other factors may be involved. Many, however, stated (wrongly) that the reason why the given statement is wrong is that the data in the table show that the progression is sometimes up and sometimes down.	Allow One may depend on another factor Allow without context NOT eg UR is independent NOT eg Only for the given years NOT eg Only for certain months
		iii	$b' = \frac{s_{xy}}{s_{yy}} = \frac{-4.624}{9.076} (= -\frac{1156}{2269} \text{ or } -0.50948)$	M1	ft their S's	If y on x: $b = \frac{s_{xy}}{s_{xx}} = \frac{-4.624}{9.241} (= -0.500)$ M1
		iii	$x - \frac{70.3}{10} = -\frac{1156}{2269} (y - \frac{30.8}{10})$	M1	or $a' = -\frac{1156}{2269} \times (-\frac{30.8}{10}) + \frac{70.3}{10}$	$y - 3.08 = -0.500 \times (x - 7.03)$ or $a = 3.08 + 0.5 \times 7.03$ M1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	$x = -0.51y + 8.6$ (2 sfs) or $x = -\frac{1156}{2269}y + 8.6$	A1	NB use b' ($= -0.509$), not r ($= -0.5049$) <u>Examiner's Comments</u> Some candidates misread the question and found the equation of the (more usual) y on x line. However, the majority read the question correctly and attempted to find the equation of the x on y line. But many of these failed to be consistent in their intention. Some found the gradient (b') correctly but then reverted to the method for y on x , attempting $a' = \bar{y} - b' \times \bar{x}$, which is incorrect. Still others found b' and a' correctly but wrote their final answer as $y = b'x + a'$ instead of $x = b'y + a'$	$y = -0.50x + 6.6$ A0
		iv	$x = -0.509 \times 4.0 + 8.60$	M1	Allow sub $y = 0.04$ for M1 only	If y on x found in (a)
		iv	$= 6.56$ (3 sf) or 6.6 (2 sf)	A1ft	ft their equn; ans to 2 sf <u>Examiner's Comments</u> This was well answered, with follow-through from incorrect equations allowed. A few candidates substituted for x instead of y .	$4.0 = -0.500x + 6.60$ M1 $x = 5.2$ (2 sf) A1ft
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
13		i	$S_{xx} = 8700000 - \frac{7000^2}{6} \quad (= 533333)$ $S_{xy} = 509900 - \frac{7000 \times 456}{6} \quad (= -22100)$	M1	Correct subst'n in any correct S formula	
		i	$b = -\frac{"22100"}{"533333"} \text{ or } -\frac{663}{16000} \quad (= -0.0414)$	M1	Correct subst'n in any correct b formula from two correct S formulae	
		i	$y - \frac{456}{6} = "-0.0414"(x - \frac{7000}{6})$	M1	ft their b except if using r	or $a = \frac{456}{6} - ("-0.0414") \times \frac{7000}{6}$ oe ft " b "
		i	$y = -0.0414x + 124 \text{ (3 sf)}$	A1	or $y = -\frac{663}{16000}x + \frac{3979}{32}$ or $y = -0.041x + 124$ Examiner's Comments This part was answered very well on the whole. A few candidates made a sign error when substituting b (which is negative) in order to find a . Some simply lost the minus sign in b . Some found b and stopped. Others found r instead of what was asked. A few found r and then used this as their value of b .	Allow $y = -0.04x + 124$ if $-0.041...$ seen
	ii		70 to 72	B1	or 71 per thousand, NOT 71000 Examiner's Comments Many candidates ignored the instruction to "use the regression line in the diagram" and used their equation from (i). A few candidates misinterpreted the situation, giving an answer such as 71 000.	No ft from (i) Ignore method



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Extrapolation oe	B1	Allow "2400 is beyond graph" } "Not shown on the graph" or } 1 st B1 only "Line drops low, or below 0" } "Outlier" }	"Line only allows for countries poorer than Nigeria" 1 st B1 Allow "Value for Nigeria is -ve 1 st B1
		iii	Corr'n not high or small sample	B1	Poor corr'n oe, or pts not close to line oe 2 nd B1 Examiner's Comments Most candidates gave one correct answer, using the word "extrapolation" or some equivalent wording. However, many either gave no second reason or gave one that was, in effect, equivalent to their first reason (eg "The IMR will become negative"). Some candidates gave the valid second reason, namely that the diagram does not show good linear correlation.	NOT "Other factors may apply" oe Ignore all else
		iv	$S_{xx} = 8700000 + 1300^2 - \frac{(7000+1300)^2}{7}$		or $10390000 - \frac{(8300)^2}{7} = \frac{3840000}{7}$ or 548571	
		iv	$S_{yy} = 36262 + 96^2 - \frac{(456+96)^2}{7}$	M1	or $45478 - \frac{552^2}{7} = \frac{13642}{7}$ or 1948.86	Correct sub in any correct S formula M1 Correct value of any S seen or implied by r A1
		iv	$S_{xy} = 509900 + 1300 \times 96 - \frac{8300 \times 552}{7}$	A1	or $634700 - \frac{8300 \times 552}{7} = -\frac{138700}{7}$ or -19814.3	
		iv	$r = \frac{-19814.3}{\sqrt{548571 \times 1948.86}}$	M1	Correct subst'n in any correct r formula from 3 correct subs in 3 correct S formulae, ie all correct method	SC If $n = 6$, but otherwise correct allow M1A0M1A0 (ans $r = -0.574$, must see wking)



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	= -0.606 (3 sf)	A1	Examiner's Comments This part was answered well by most candidates. A few used the original totals, just changing the value of n from 6 to 7. Others made the opposite error, finding new totals, but with $n = 6$. Sensibly, most wrote down their new totals such as $\sum xy$, but some were incorrect and, without any indication of method, these lost a method mark.	
		v	No effect oe	B1	Stay the same oe Allow just "No" Examiner's Comments A few candidates thought that r would decrease because the values used in the formula would decrease, but most stated correctly that r would be unchanged.	Ignore all else
			Total	12		



Question			Answer/Indicative content	Marks	Part marks and guidance	
14		i	$(a) S_{xx} = 3215 - \frac{179^2}{12}$ $(\text{= } 544.9167 \text{ or } \frac{6539}{12})$ $S_{yy} = 3565 - \frac{191^2}{12}$ $(\text{= } 524.9167 \text{ or } \frac{6299}{12})$ $S_{xy} = 3343 - \frac{179 \times 191}{12}$ $(\text{= } 493.9167 \text{ or } \frac{5927}{12})$	M1	Correct sub in a correct S formula	
		i	$r = \frac{"493.9167"}{\sqrt{"544.9167" \times "524.9167"}}$	M1	Correct sub in 3 correct S formulae and a correct r formula	Correct ans, (≥ 3 sf) no wking, M1M1A1
		i	= 0.924 (3 sf) or 0.9235.....	A1	Must see ≥ 3 sf	Ignore any comparison with 0.92 Examiner's Comments The vast majority of candidates answered this well, with the actual calculation being carried out accurately in most cases. A few truncated their answer to 3 significant figures, rather than rounding it, thereby losing a mark. There were very few cases of premature rounding of one or more S-values.
		i	(b) Correlation does not imply causation or It depends on the time of year or Both depend on a third variable or There could be other factors	B1	Any answer which implies or is equiv to any of these	Allow without context. Ignore incorrect comments Examiner's Comments This is a simple, standard, and very common, question. Nevertheless, many candidates appeared not to be aware of the fact that correlation does not imply causation.



Question			Answer/Indicative content	Marks	Part marks and guidance	
	ii		(a) 'Increased' and Positive gradient or positive coeff of n or 'Output goes up by 0.6 each month' Both needed	B1	'Increased' and values of z shown as follows: at least 6 values or 1st and last values or 1st, or 2nd or 3rd or 4th and 9th or 10th or 11th or 12th ie 17.6 or 18.2 or 18.8 or 19.4 and 22.4 or 23 or 23.6 or 24.2	'Increased' and 'Value of $0.6n$ increases as n increases' NOT: 'Increased' and 'Value of z incr as n incr' 'z incr as no. of mths incr Examiner's Comments Disappointingly few candidates gave the simple answer that the gradient of the regression line is positive and therefore the output increases. The majority actually calculated some of the values of the output. Some even calculated all 12 values! Many gave only the first two or three values, but this did not gain the mark because it does not comprise sufficient evidence that the output generally increased. In parts (ii)(b) and (ii)(c) many candidates used the unnecessarily long methods described below. They gained full marks if they obtained the correct answers, but the large amount of arithmetic involved meant that they were liable to make arithmetical errors.
	ii		(b) $\bar{n} = 6.5$ or $\frac{78}{12}$ oe seen	B1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	$\bar{z} = 0.6 \times '6.5' + 17$ alone, eg not $\div 12$ or $17 = \bar{z} - 0.6 \times "6.5"$ oe	M1	or $(0.6 \times 1 + 17 + 0.6 \times 2 + 17 + \dots 0.6 \times 12 + 17) \div 12$ oe or $'250.8' \div 12$ M1 ft their '6.5' only if comes from $\dots \div 12$	Long method, all correct terms seen and $\div 12$ M1 NB ans 20.9 may not score the B1
		ii	$\bar{z} = 20.9$	A1	cao	Examiner's Comments Most candidates found $\bar{n}$ but, again disappointingly, many found $\bar{z}$ by using the equation of the regression line to find all twelve values of z rather than by substituting their value of $\bar{n}$
		ii	(c) Total output = "20.9" $\times$ 12	M1	or $0.6 \times 1 + 17 + 0.6 \times 2 + 17 + \dots 0.6 \times 12 + 17$ oe or eg $\frac{88}{5} + \frac{91}{5} + \dots + \frac{121}{5}$ oe	Long method, all correct terms seen
		ii	251 (3 sf)	A1f	ft their (ii)(b)	Not ISW, eg 25100 scores A0, even if 251 seen Examiner's Comments Again, many candidates worked out all 12 values of the output and averaged them, rather than multiplying their answer to part (ii)(b) by 12.
			Total	10		



Question			Answer/Indicative content	Marks	Part marks and guidance	
15		i	$S_{xx} = 200.39 - \frac{33.7^2}{6}$		or 11.108 or 11.1 or $\frac{1333}{120}$	
		i	$S_{yy} = 28314 - \frac{410^2}{6}$		or 297.333 or 297 or $\frac{892}{3}$	
		i	$S_{xy} = 2313.9 - \frac{33.7 \times 410}{6}$	M1	or 11.067 or 11.1 or $\frac{166}{15}$	Correct sub in a correct S formula or correct value of one S seen
		i	$r = \frac{"11.067"}{\sqrt{"11.108" \times "297.333"}}$ = 0.193 (3 sf)	M1		Correct sub in 3 correct S formulae and a correct r formula
		i		A1	Examiner's Comments This question was well-answered, with only a few candidates showing some of the errors mentioned above, including giving only an answer correct to two significant figures, rather than the standard three.	No working: 0.193 M1M1A1 Ignore comment about $0 < r < 0.2$

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	(For these 6 clubs) No / little / poor / weak oe Relationship / corr'n / link oe between (top) salaries and no. of points	B1	Allow without "For these 6 clubs" & "top" or "no strong corr'n between etc" In context. Examiner's Comments This was also well-answered, although a significant minority lost the mark by not setting their answer in context. Some others lost the mark by referring to goals instead of points. There were some imaginative answers such as 'The value of r shows whether the players were worth what they got paid' but these did not gain the mark. Some candidates failed to understand the significance of the small value of r , giving answers such as 'There is a small correlation between the top salary and points scored. As the top salary increases, the points scored increases.'	Allow "Salary has little effect on points" Ignore all else including "positive" NOT if use "goals" instead of "points"

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		iii	Extrapolation oe Corr'n poor / weak or no rel'nship / link oe or Points not close to line Small sample or only (top) 6 clubs oe	B1B1	Outside range of values. Salary is less than the others. r small or r close to 0 or r not close to 1 or Results do not correlate well Any two; allow without context Examiner's Comments Most candidates gave at least one of the two rote answers (Extrapolation and poor correlation), but many gave two answers that amounted to the same thing. Some gave the rote answer 'Correlation does not imply causation', which did not gain a mark.	NOT "Corr'n does not imply causation" NOT "Could be other factors" NOT if use "goals" instead of "points"
			Total	6		



Question			Answer/Indicative content	Marks	Part marks and guidance	
16		i	$S_{xx} = 481.13 - \frac{60.5^2}{8}$ or 23.59875 or 23.6 or $\frac{18879}{800}$ $S_{xy} = 334.65 - \frac{60.5 \times 44.9}{8}$			Alternative method: $44.9 = 8a + 60.5b \text{ M1}$ $334.65 = 60.5a + 481.13b \text{ M1}$ hence $a = 7.18$ or $b = -0.208$ A1 $y = -0.208x + 7.18 \text{ A1}$
		i	or - 4.90625 or - 4.91 or $-\frac{157}{32}$ $b = \frac{334.65 - \frac{60.5 \times 44.9}{8}}{481.13 - \frac{60.5^2}{8}} \text{ oe}$	M1	Correct sub in any correct S_{xx} or S_{xy} formula or correct value of either S	
		i	or - 0.20790 or - 0.208 or $-\frac{3925}{18879}$	M1	Correct sub in both Ss and in a correct b formula	
		i	$y - \frac{44.9}{8} = "-0.20790"(x - \frac{60.5}{8})$ $y = -0.208x + 7.18$ (or + 7.19) (3 sf)	M1 A1	or $a = \frac{44.9}{8} - "-0.20790" \times \frac{60.5}{8}$ or $y = -\frac{3925}{18879}x + 7.18/9$ Must include "$y =$" Allow $y = -0.21x + 7.2$ (awrt 2 sf) Examiner's Comments This question was very well-answered. Occasionally the errors mentioned above were seen and sign errors sometimes crept in.	no wking, correct ans M1M1M1A1 If find x on y line, can score first M1 only or ans $x = 31 - 4.2y$ seen first M1 only
		ii	"- 0.208" $\times$ 9.2 + "7.18"	M1		



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	= 5.27 or 5.28 (km/l) (3 sf)	A1ft	ft their equn from (i) Examiner's Comments Most candidates scored full marks on this question.	but no ft from x on y line
		iii	(7.56, 5.61) (3 sf) or $(\frac{121}{16}, \frac{449}{80})$ oe	B1	Ignore calc'n of reg line, if done Examiner's Comments Some candidates knew what was required, but many gave answers that suggested that they did not understand that the two regression lines both pass through the point $(\bar{x}, \bar{y})$. The most common incorrect answer was (0, 0). Some candidates gave the answer 'The midpoints of the two lines'. A common answer was (0, 7.18). Some candidates just quoted one or two of the points given in the table. Some substituted $y = x$ into the equation found in part (i) and solved to find x and y. Many gave coordinates without any indication of how they had been obtained.	NOT $(\frac{60.5}{8}, \frac{44.9}{8})$
		iv	Use reg line of x on y (either equn or line)	M1	Must specify or imply x on y, otherwise M0A0 NOT "Use either x on y or y on x"	If calc x on y reg line (allow errors) M1



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	Sub $y = 5.8$ or fuel = 5.8 or km/l = 5.8	A1	NOT “and read off v coord” Examiner’s Comments The most frequent error was not recognising that the x on y regression line is required, and many incorrect answers began ‘Rearrange the equation ...’. A few candidates correctly stated that the x on y line must be used but omitted to say that $y = 5.8$ needs to be substituted into the equation or read off from the line. Some candidates mentioned a non-specific line of best fit.	Subst 5.8 into their x on y line A1 Ignore all else
			Total	9		



Question			Answer/Indicative content	Marks	Part marks and guidance	
17		i	$S_{xx} = 20400 - \frac{360^2}{8} \quad (= 4200)$			
			$S_{yy} = 6.88 - \frac{6.8^2}{8} \quad (= 1.1)$			
		i	$S_{xy} = 241 - \frac{360 \times 6.8}{8} \quad (= -65)$	M1	Correct sub in a correct S formula	
		i	$r = \frac{"-65"}{\sqrt{"4200" \times "1.1"}}$	M1	Correct sub in 3 correct S formulae and a correct r formula	
		i	= -0.956 (3 sf)	A1	Correct ans with no working M2A1	Ignore comment about $-1 < r < -0.9$
					Examiner's Comments This part was well answered by most candidates. A few made the errors mentioned above. A significant number made an arithmetical slip leading to $S_{xx} = -66$ instead of -65. Over rounding was frequently seen in this question, i.e. '0.96'.	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		ii	eg As you move further away, prices drop	B1	High prices go with short distances oe Allow "Strong (or high or good or equiv) neg corr'n between price and distance" Examiner's Comments Many candidates failed to mention the context. Others stated that the value of r shows "negative correlation" between price and distance, omitting to include "strong" or an equivalent. A few candidates simply discussed the nearness of points to a line of regression. Some candidates thought that a high negative value of r means that there is very little correlation.	Both variables must be in context; miles & £ enough Ignore all else, even if incorrect NOT just neg corr'n between price & dist
		iii	None	B1	Examiner's Comments Most candidates answered this part correctly, although some did not appreciate the difference between "affect" and "effect". They were not penalised for this non-mathematical error.	Ignore all else, even if incorrect
		iv	$b = \frac{-65}{4200}$ ($= -0.0154762$)	M1	ft their S_{xy} & S_{xx} from (i) for M-marks only	or fresh start correct method
		iv	$Y - \frac{6.8}{8} = -0.0154762(x - \frac{360}{8})$ oe	M1	or $a = \frac{6.8}{8} + "0.0154762" \times \frac{360}{8}$ oe	



Question			Answer/Indicative content	Marks	Part marks and guidance	
		iv	$y = -0.0155x + 1.55$ (3 sf) oe or $y = \frac{433}{280} - \frac{13}{840}x$ oe	A1	allow $y = -0.015x + 1.5$ (or figs which round to these) (NOT $y = -0.016x + 1.6$ NOT $y = -0.02x + 1.5$) Correct ans with no working M2A1 Examiner's Comments Like part (i) this was generally well answered. Some candidates started from scratch, not appreciating that they had already found the values of S_{xx} and S_{xy} in part (i). A few of these, having found the S values correctly in part (i), tried to use the formula for b from MF1 ($b =$ $\frac{\sum(x-\bar{x})(y-\bar{y})}{\sum(x-\bar{x})^2}$) which they clearly did not understand. Some used incorrect values such as $S_{xy} = 241$. Others made errors in the sign or the decimal point or rounding. Subtracting a negative to find 'a' caused problems for some candidates.	Must have "y = " Allow figures in equn which round to the correct figures to either 3 sf or 2 sf, Correct ans with no working M2A1 even if they result from arith errors.

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Question			Answer/Indicative content	Marks	Part marks and guidance	
		v	Values of x are chosen beforehand or x is independent or controlled	B1	x is fixed or given or set or predetermined or Examiner's Comments Many candidates answered correctly. A few gave inadequate statements, such as that y is dependent on x . Others gave incorrect statements such as that y is independent or that x depends on y . Some wrote "because r is close to -1 ". A large number stated that this would be a better method because no rearrangement would be necessary.	Not " x is constant." Not just " y depends on x " Ignore all other, even if incorrect
			Total	9		

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